Direction Instability Predicts Cross-Cell-Line Drug Mechanism Transport in LINCS L1000

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Abstract

A drug's mechanism of action may or may not generalize across cell types. We adapt a geometric measure—direction instability, the mean pairwise angular deviation of perturbation signature vectors—to quantify whether a drug's transcriptomic effect is conserved across biological contexts. Applied to 8,949 compounds tested in ≥ 5 cell lines from the LINCS L1000 Connectivity Map (978 landmark genes, 167,266 signatures), direction instability separates mechanism-conserving drugs from context-dependent ones (population-null $p < 0.01$ for 78 drugs). Direction instability anticorrelates with gene-level Jaccard overlap of top perturbed genes (Spearman $\rho = -0.79$), indicating that directional change reflects biological mechanism change. Within the HDAC inhibitor class, pan-isoform inhibitors transport across cell lines (mean instability 0.62, $n = 8$) while isoform-selective inhibitors do not (mean instability 0.96, $n = 5$), showing that target breadth—not drug class membership—determines cross-context transcriptomic generalizability. Genetic triangulation against 154,993 shRNA knockdown signatures failed to reach the pre-registered effect-size threshold: the global correlation was statistically significant but small ($\rho = -0.18$, $p = 1.1 \times 10^{-16}$; threshold $|\rho| > 0.3$). Exploratory within-family stratification yielded stronger effects ($|\rho| = 0.32$–0.84 across five of 54 families, none surviving Bonferroni correction), a result that requires independent replication. Leave-one-out cross-validation across 2,694 drugs shows that instability computed on $N-1$ cell lines predicts whether the held-out cell line will show a consistent response (AUROC = 0.986; permutation control collapses to 0.500 ± 0.012, indicating that the signal reflects genuine drug-level structure despite partial geometric circularity between predictor and outcome).

1 Introduction

Pharmacological perturbation experiments increasingly rely on profiling drug effects across panels of cell lines to identify compounds with robust mechanisms of action (Lamb et al., 2006; Subramanian et al., 2017). The LINCS L1000 Connectivity Map measures 978 landmark gene expression changes for ∼20,000 compounds across up to 70 cell lines, producing high-dimensional signature vectors that characterize each drug's transcriptomic effect in each context (Subramanian et al., 2017).

A fundamental question arises: when does a drug's transcriptomic effect *transport* from one cell type to another—that is, when does the drug perturb the same genes, in the same directions and relative proportions, regardless of cellular context? Two drugs may share a mechanism-of-action label yet differ in how consistently they perturb gene expression across contexts. A pan-HDAC inhibitor like vorinostat affects chromatin remodeling in every cell, while an HDAC8-selective inhibitor like PCI-34051 only produces a strong transcriptomic signature where HDAC8 is functionally relevant.

We formalize this distinction using **direction instability**: the mean pairwise angular deviation of a drug's unit-normalized signature vectors across cell lines. This metric is adapted from the neural bracket norm (Tower, 2026), which quantifies how much a learned representation's direction rotates across evidence levels. Here, the analogy maps evidence quartiles to cell lines, and neural activation vectors to drug perturbation signatures.

Direction instability decomposes the perturbation effect into two independent components. The *direction* captures which genes are perturbed and in what relative proportions. The *magnitude* captures the overall effect size. A drug with low direction instability but high magnitude variation has a conserved qualitative mechanism with context-dependent penetrance—a biologically meaningful and pharmacologically useful category.

We apply this framework to 8,949 drugs with ≥ 5 cell lines in LINCS L1000. Because a drug's direction instability depends on its cell-line count, we introduce a coverage-conditioned population null that compares each drug against references with similar count, providing a reusable significance test applicable to any multi-context profiling dataset. The scorer and validation labels were frozen under a pre-registered commit SHA (`1dc20a2`) before any real data were analyzed.

2 Methods

2.1 Data

Perturbation signatures were drawn from the LINCS L1000 Phase I dataset (GEO accession GSE92742), using Level 5 replicate-collapsed z-scores across 978 landmark genes (Subramanian et al., 2017). Metadata (signature identifiers, perturbation types, cell line annotations) were downloaded from GEO and filtered to small-molecule perturbagens (`pert_type = trt_cp`) with ≥ 5 distinct cell lines. Replicate signatures within a drug–cell-line pair were averaged to yield one consensus signature per drug per cell line.

Genetic perturbation signatures were drawn from the same dataset by filtering to shRNA knockdown perturbagens (`pert_type = trt_sh`), yielding 154,993 signatures across 4,371 gene targets and 20 cell lines. Drug-target annotations from the Repurposing Hub mapped 1,067 drugs to gene targets present in the shRNA data.

Mechanism-of-action (MOA) annotations were obtained from the Broad Drug Repurposing Hub (version 2020-03-24) (Corsello et al., 2017), providing MOA labels for 1,782 of the 8,949 drugs. These labels were frozen before analysis as part of the pre-registration commit (SHA `1dc20a2`; Section 2.5).

2.2 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \ldots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define the unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

$$D = 1 - \frac{2}{K(K-1)} \sum_{i<j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \tag{1}$$

$D \in [0, 2]$. $D = 0$ when all signatures point in the same direction. $D = 1$ when signatures are orthogonal on average. $D > 1$ indicates anticorrelation.

2.3 Complementary metrics

Magnitude CV: coefficient of variation of signature L_2 norms $\{\|\mathbf{s}_i\|\}_{i=1}^{K}$, capturing effect-size variation independent of direction.

Fréchet variance: mean squared geodesic deviation from the Fréchet mean on the unit sphere S^{977} (Pennec, 2006).

Gene-level Jaccard overlap: for each pair of cell lines, the Jaccard index of the top-50 up-regulated and top-50 down-regulated genes, averaged over all pairs and both directions.

2.4 Population null model

A drug's direction instability depends on its cell-line count: drugs tested in more cell lines have more pairwise comparisons and tend toward higher baseline instability. To control for this confound, we compare each drug against a reference population of drugs with similar coverage: all drugs with cell-line count in $[K/2,\ 2K]$, where K is the focal drug's count. The p-value is the fraction of reference drugs with instability $\leq$ the focal drug's instability. This replaces a permutation null, which is invariant under cell-line label shuffling for pairwise cosine statistics.

2.5 Pre-registration

The scoring functions (Equation 1 and all complementary metrics), the frozen MOA labels, and the pre-registered hypotheses were committed together in a single commit before any real LINCS signatures were analyzed (commit SHA `1dc20a2`; scorer in `geometry/drug_transport.py`, frozen labels in `data/frozen_drug_labels.json`, hypotheses in `PREREGISTRATION.md`).

3 Results

3.1 Direction instability tracks biological mechanism conservation

Landscape. Among 8,949 drugs with ≥ 5 cell lines, mean direction instability was 0.924 ± 0.061 (median 0.940). Most drugs exhibit context-dependent effects. Only 78 drugs (0.9%) achieved $p < 0.01$ under the population null, and 411 (4.6%) achieved $p < 0.05$, indicating that genuinely mechanism-conserving drugs are rare.

Cell-line coverage ranged from 5 to 64 (mean 10.2, median 8). The population null accounts for this variation: a drug's instability is evaluated against drugs with comparable coverage.

Gene-level validation. Direction instability anticorrelated with gene-level Jaccard overlap of top perturbed genes: Spearman $\rho = -0.79$ ($p < 10^{-300}$; Figure 1). Drugs with low instability perturb the same genes across cell types, while high-instability drugs engage different gene programs in different contexts. HDAC inhibitors trace the selectivity gradient along this curve: pan-isoform inhibitors cluster at low instability and high Jaccard overlap, while isoform-selective inhibitors are indistinguishable from the bulk population.

Direction instability and magnitude CV are partially dissociable (Spearman $\rho = -0.10$, $p = 2.1 \times 10^{-21}$): 7.8% of drugs fall in the low-direction, high-magnitude quadrant, indicating conserved mechanism with context-dependent penetrance.

MOA class stratification. Figure 2 shows mean direction instability by MOA class (classes with ≥ 15 annotated drugs). Two classes stand out: HDAC inhibitors ($\bar{D} = 0.77$, $\sigma = 0.14$) and topoisomerase inhibitors ($\bar{D} = 0.77$, $\sigma = 0.12$). CDK inhibitors and tubulin polymerization inhibitors also show reduced instability ($\bar{D} = 0.80$ and 0.88 respectively). Receptor antagonists and agonists cluster near the population mean ($\bar{D} = 0.93$–0.96), consistent with receptor-mediated mechanisms being cell-type-dependent.

3.2 Target breadth determines transport

The HDAC selectivity gradient. The HDAC inhibitor class provides a natural experiment: 20 annotated HDAC inhibitors span the full selectivity spectrum from pan-isoform to isoform-specific (Figure 3). Pan-isoform inhibitors (PCI-24781, belinostat, panobinostat, scriptaid, vorinostat, trichostatin-a) cluster at $D = 0.56$–0.67 with significant population p-values ($p < 0.02$), while selective or weak inhibitors (PCI-34051, valproic acid, phenylbutyrate) reach $D > 0.95$ and are indistinguishable from the null. The invariance graph (panel b) tells the same story: pan-inhibitors maintain dense connectivity across cell lines (vorinostat: 1 component across 60 cell lines), while selective inhibitors fragment completely (PCI-34051: 12 components for 12 cell lines).

Pan-HDAC inhibitors (mean $D = 0.62$, $n = 8$) transport their mechanism across cell lines because their targets—histone deacetylases of classes I, II, and IV—are expressed in every cell type (Bolden et al., 2006). Chromatin remodeling is a universal process, so inhibiting it broadly produces a conserved transcriptomic signature.

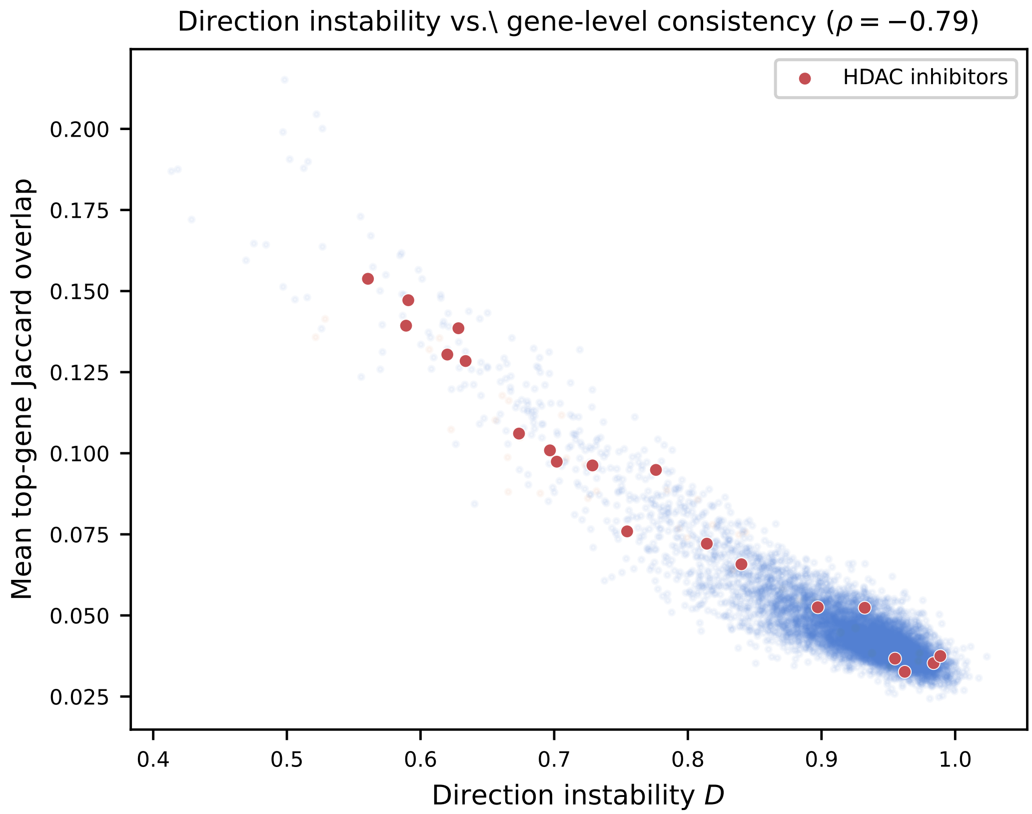

Figure 1: Direction instability versus gene-level consistency (mean top-gene Jaccard overlap) for 8,949 drugs. HDAC inhibitors are highlighted in red. The strong anticorrelation ($\rho = -0.79$) indicates that directional change reflects which genes are perturbed, not overall effect size. HDAC inhibitors span the gradient from pan-isoform (low D, high Jaccard) to isoform-selective (high D, low Jaccard).

Selective HDAC inhibitors (mean $D = 0.96$, $n = 5$) do not transport. PCI-34051 targets HDAC8 specifically (Balasubramanian et al., 2008); its transcriptomic effect depends on whether HDAC8 is functionally relevant in a given cell type. Valproic acid and phenylbutyrate are weak HDAC inhibitors with additional pharmacological activities, so their signatures reflect cell-type-specific off-target effects rather than HDAC inhibition per se.

Entinostat ($D = 0.70$, class I–selective) falls between the two groups, consistent with intermediate target breadth: class I HDACs (HDACs 1, 2, 3) are more widely expressed than HDAC8 but narrower than the full isoform panel.

Top transporting drugs. The 10 most mechanism-conserving drugs with MOA annotations target processes common to all cells (Table 1): protein synthesis (homoharringtonine), RNA transcription (dactinomycin), DNA topology (epirubicin, doxorubicin), chromatin remodeling (PCI-24781, belinostat, panobinostat), cell cycle progression (BMS-387032), and signal transduction kinases (bisindolylmaleimide-IX). These mechanisms operate independently of tissue-specific receptor expression, explaining their cross-context conservation.

Conversely, imatinib ($D = 0.97$, 19 cell lines) exemplifies a context-dependent drug (Deininger et al., 2005). Its primary therapeutic target is BCR-ABL, but it hits eight kinase targets (ABL1, KIT, PDGFRA/B, DDR1, RET, NTRK1, CSF1R). Across the LINCS panel, its signatures rotate 88° on average—near orthogonality. Genetic triangulation reveals the mechanism: imatinib's highest drug–shRNA cosine is with DDR1 ($\cos = 0.22$), not ABL1 ($\cos = 0.18$), because most LINCS cell lines lack the Philadelphia chromosome. In

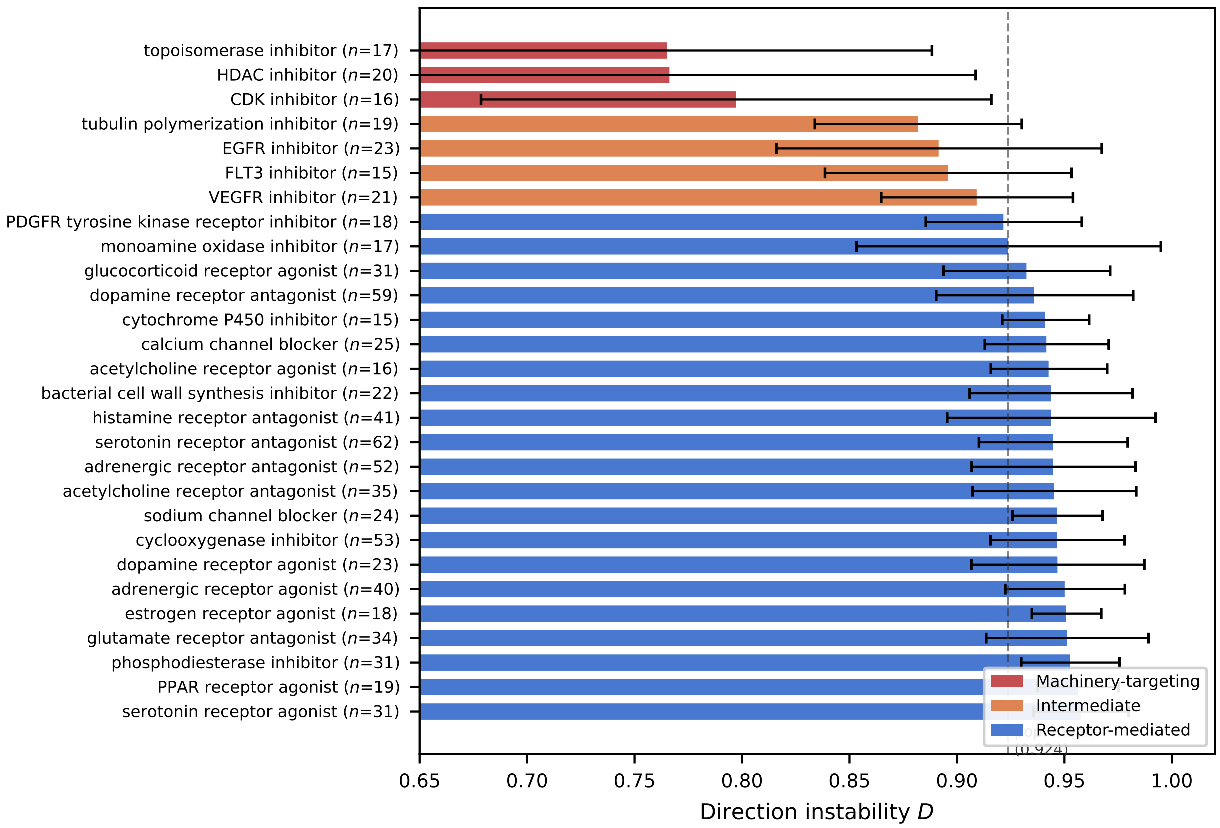

Figure 2: Direction instability by MOA class ($n \geq 15$). Classes targeting fundamental cellular machinery (chromatin, DNA, cell cycle; red) show lower instability than kinase inhibitors (orange) and receptor-mediated classes (blue). Dashed line: population mean ($D = 0.924$). Error bars: ± 1 SD. Within HDAC inhibitors, the high variance reflects the pan-selective gradient (Figure 3).

Table 1: Top 10 transporting drugs with MOA annotation.

Drug	MOA	D	p
homoharringtonine	protein synthesis inhibitor	0.497	0.001
dactinomycin	RNA polymerase inhibitor	0.506	0.002
BMS-387032	CDK / cell cycle inhibitor	0.522	0.002
bisindolylmaleimide-IX	PKC inhibitor	0.526	0.003
epirubicin	topoisomerase inhibitor	0.529	0.003
anisomycin	DNA synthesis inhibitor	0.555	0.004
PCI-24781	HDAC inhibitor	0.561	0.004
BNTX	opioid receptor antagonist	0.564	0.003
digitoxigenin	ATPase inhibitor	0.586	0.004
belinostat	HDAC inhibitor	0.589	0.006

each cell line, a different off-target kinase dominates the transcriptomic response. The population null correctly identifies this instability as unremarkable ($p = 0.80$): imatinib's context dependence is typical among drugs tested in 19 cell lines. In the held-out prediction analysis (Section 3.3), a screening pipeline treating

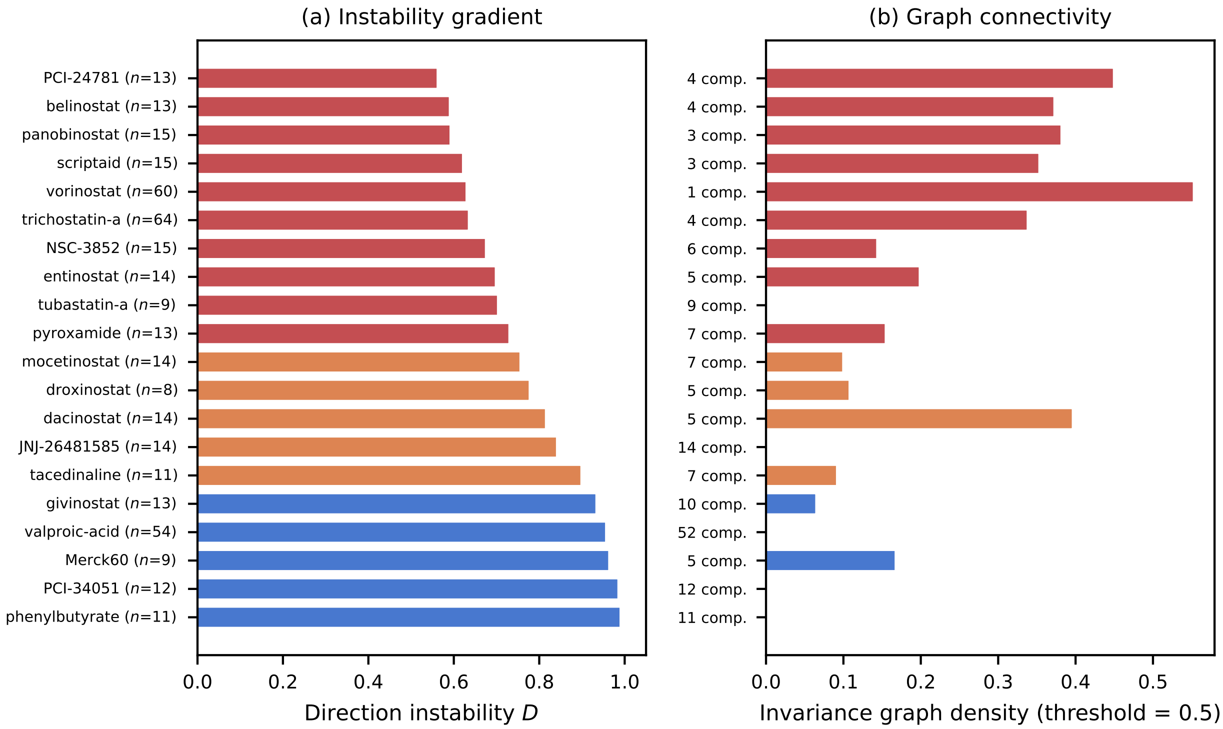

Figure 3: HDAC inhibitor selectivity gradient. **(a)** Direction instability for all 20 HDAC inhibitors, ordered by D. Pan-isoform inhibitors (red) cluster at low instability; selective and weak inhibitors (blue) are indistinguishable from the population. **(b)** Invariance graph density at cosine threshold 0.5 and number of connected components. Pan-inhibitors form few components (dense graphs); selective inhibitors fragment into isolated nodes.

imatinib's LINCS signature as a portable biomarker would draw wrong conclusions in 42% of new contexts (held-out consistency: 58%).

Cytotoxicity defense. A potential confound: drugs that kill cells uniformly might produce conserved signatures through a generic stress response rather than target-specific mechanism transport. We tested this by computing the mean cytotoxic signature across known cytotoxic drugs (topoisomerase inhibitors, DNA alkylating agents, tubulin inhibitors, proteasome inhibitors, RNA polymerase inhibitors, protein synthesis inhibitors) and measuring each drug's cosine similarity with this reference.

Direction instability showed weak correlation with cytotoxic-signature overlap (Spearman $\rho = -0.17$, excluding known cytotoxic drugs; Figure 4a). The correlation is in the wrong direction for the confound hypothesis: drugs resembling the cytotoxic profile are *less* stable, not more. Removing the 50 genes most associated with the cytotoxic signature and recomputing direction instability preserved the HDAC inhibitor gradient ($\rho = 0.83$ between original and stress-corrected values; Figure 4b). Pan-HDAC inhibitors became *more* stable after stress-gene removal (vorinostat: $D = 0.63 \to 0.50$), confirming that their conserved signal is chromatin-specific, not a generic stress response.

The transporting core. For each drug, we defined a "transporting core": genes with consistent sign (>80% of cell lines agree), low magnitude CV (<1.0), and meaningful effect size (mean $|z| > 0.5$). Core size anticorrelated with direction instability (Spearman $\rho = 0.69$, $p < 10^{-300}$), indicating that transporting drugs affect a larger set of genes consistently.

The 8 pan-HDAC inhibitors shared a 26-gene core—genes consistently perturbed by all 8 drugs across all cell lines. This shared core includes *SUV39H1* (a histone methyltransferase directly involved in chromatin

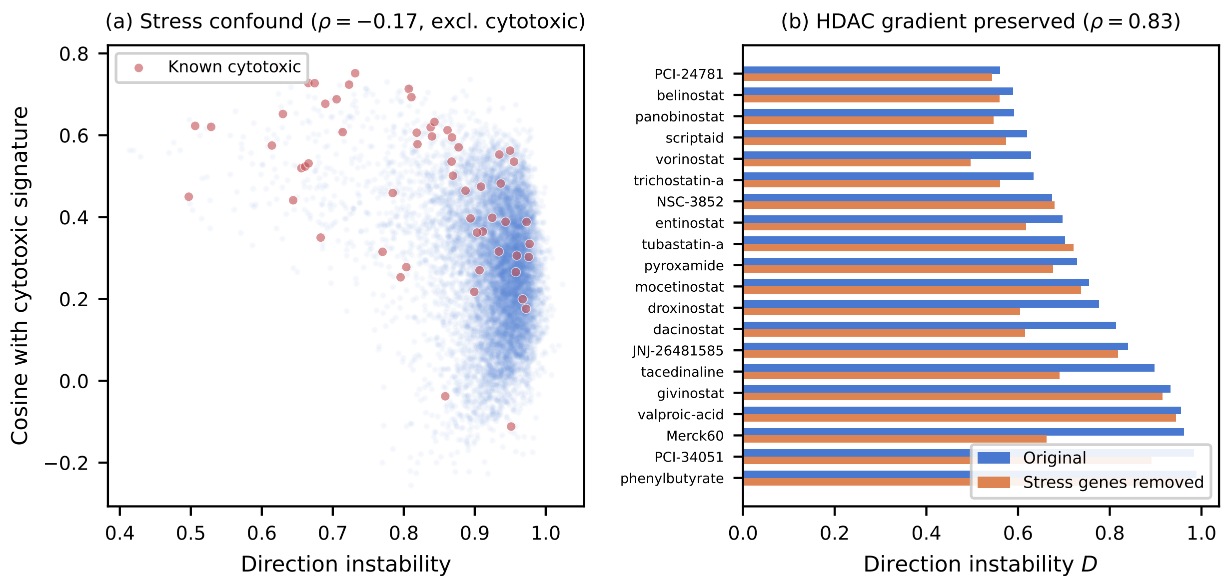

Figure 4: Cytotoxicity confound check. **(a)** Direction instability versus cosine similarity with the mean cytotoxic signature ($\rho = -0.17$, excluding known cytotoxic drugs in red). The negative sign rules out the confound: drugs resembling the stress response are less stable, not more. **(b)** HDAC inhibitor instability before (blue) and after (orange) removing 50 stress-associated genes. The selectivity gradient is preserved ($\rho = 0.83$), and pan-inhibitors become more stable.

remodeling), *MYC* (a master transcription factor regulated by HDAC activity), *CDK6* (cell cycle kinase transcriptionally repressed by HDAC inhibitors), *BIRC5* (survivin, a known HDAC-responsive anti-apoptotic gene), and *ORC1* (DNA replication origin licensing). The transporting core is the target pathway: pan-HDAC inhibitors share a chromatin/cell-cycle gene program that operates identically across cell types.

Genetic triangulation. The preceding analyses establish that direction instability tracks mechanism conservation, but they rely on drug signatures alone. We tested a stronger prediction: if a transporting drug genuinely acts through its annotated target, its perturbation signature should resemble the transcriptomic effect of genetically knocking down that target. We computed cosine similarity between each drug's consensus signature and the shRNA knockdown signature of its annotated target gene, matched within the same cell line.

Among 2,138 drug–target pairs with ≥ 3 shared cell lines, the pre-registered global correlation between direction instability and |drug–shRNA cosine| was significant (Spearman $\rho = -0.18$, $p = 1.1 \times 10^{-16}$; Figure 5) and in the expected direction, but fell below our pre-registered effect-size threshold of $|\rho| > 0.3$.

A structural limitation of the pooled test is that it conflates drug families with inherently different baseline drug–knockdown cosines. Proteasome inhibitors produce signatures that naturally overlap with proteasome-subunit knockdown (cos ≈ 0.45), while HDAC inhibitors produce lower cosines with individual HDAC knockdowns (cos ≈ 0.15), irrespective of instability. Pooling across families introduces variance that is orthogonal to the instability signal.

We therefore performed an exploratory within-family stratification, which was not pre-registered (Table 2). Of 54 families with ≥ 15 drug–target pairs, five exceeded $|\rho| > 0.3$ at nominal $p < 0.05$. Under the null hypothesis of no association, $54 \times 0.05 = 2.7$ families would exceed this threshold by chance; observing 5 is suggestive but not definitive. None survive Bonferroni correction ($\alpha_{\text{adj}} = 0.05/54 = 0.0009$); the strongest (adrenergic receptor agonist, $p = 0.001$) is marginal. The expected sign held in all five: lower instability predicted higher drug–target cosine. Per-target-gene analysis showed convergent effects: KDR (VEGFR,

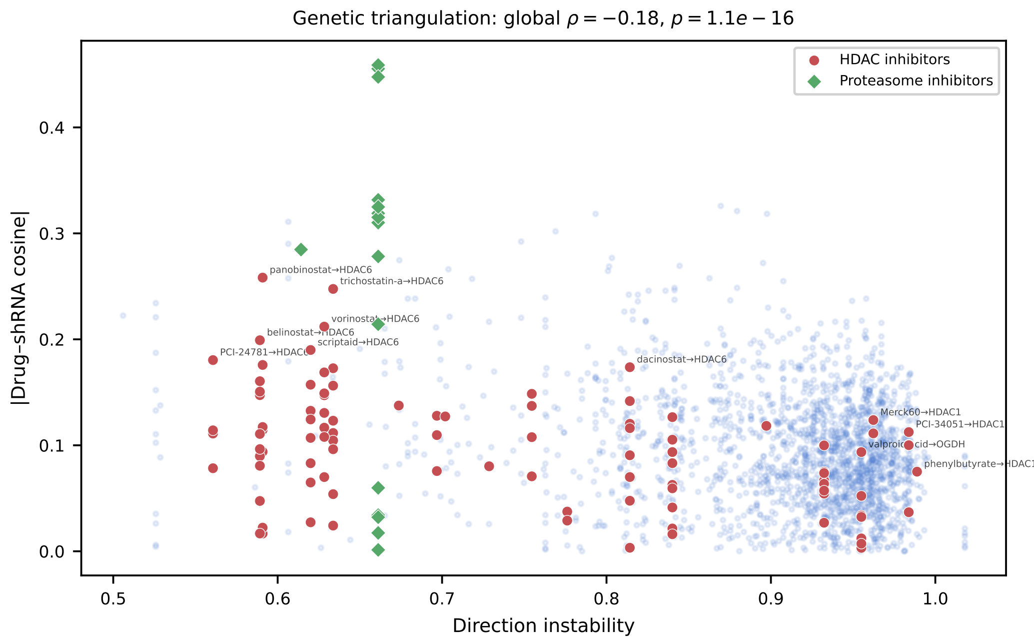

Figure 5: Genetic triangulation: direction instability versus |drug–shRNA cosine| for 2,138 drug–target pairs. HDAC inhibitors (red circles) and proteasome inhibitors (green diamonds) are highlighted. Proteasome inhibitors cluster at high cosine and low instability; HDAC inhibitors span the full gradient. Labels show drug–target pairs for selected HDAC inhibitors. The global correlation ($\rho = -0.18$) is attenuated by cross-family baseline differences (proteasome cos ≈ 0.45 vs. HDAC cos ≈ 0.15); within-family stratification yields $|\rho| = 0.32$–0.84 (Table 2).

Table 2: Exploratory within-family genetic triangulation (not pre-registered). Spearman correlation between direction instability and |drug–shRNA cosine| for each drug's annotated target, among families with ≥ 15 drug–target pairs and nominal $p < 0.10$ (5 of 54 families tested). None survive Bonferroni correction. Negative ρ indicates that mechanism-conserving drugs (low D) produce signatures more similar to genetic knockdown of their target.

MOA family	n	ρ	p
PI3K inhibitor	18	−0.55	0.017
adrenergic receptor agonist	50	−0.45	0.001
topoisomerase inhibitor	18	−0.48	0.045
EGFR inhibitor	27	−0.40	0.038
HDAC inhibitor	93	−0.32	0.002
Global (all families pooled)	2,138	−0.18	1.1×10^{-16}

$\rho = -0.84$, $p = 5.1 \times 10^{-7}$), CDK1 ($\rho = -0.79$, $p = 0.002$), AKT1 ($\rho = -0.75$, $p = 0.002$), and HDAC6 ($\rho = -0.71$, $p = 0.009$). These within-family results are exploratory and require independent replication.

The HDAC family again provides the most granular view. Among 100 HDAC drug–target pairs, pan-isoform inhibitors (PCI-24781, belinostat, panobinostat, vorinostat) showed mean |cosine| of 0.12–0.18 against their HDAC targets. PCI-34051 (HDAC8-selective, $D = 0.98$) achieved only 0.037 cosine with HDAC8—its putative primary target—and 0.10 against off-target HDACs. The selectivity gradient that appeared in the direction-instability ranking reappears in the genetic triangulation: pan-inhibitors produce transcriptomic effects that recapitulate target knockdown; selective inhibitors do not.

3.3 Predictive validation

Pre-registered hits and misses. We pre-registered the invariance-graph analysis before running it: for each drug, we built a graph over cell lines (edge = cosine similarity ≥ 0.5) and examined connected components. We predicted that pan-HDAC inhibitors would form 1–2 components while selective inhibitors would fragment completely. The ordinal gradient held: vorinostat formed 1 component across all 60 cell lines; PCI-34051, tubastatin-a, phenylbutyrate, and JNJ-26481585 each fragmented into $K = n$ components (every cell line isolated). The strict threshold did not hold: pan-HDAC inhibitors formed 1–5 components rather than the predicted 1–2 (e.g., PCI-24781 had 4 components across 13 cell lines). The gradient is unmistakable; the pre-registered threshold was too strict.

We also predicted that connected components would correspond to tissue lineages. Only 1 of 163 drugs with 2–5 components showed significant tissue enrichment in any component (Fisher's exact test, $p < 0.05$), a clear miss on the pre-registered prediction. This null is itself informative: invariance-graph structure reflects mechanism-level conservation, not tissue-of-origin expression clustering. Drugs that transport do so because of their target pathway, not because of shared lineage among responding cell lines.

Leave-one-out cross-validation. The preceding analyses show that direction instability *correlates with* various measures of mechanism conservation, but correlation does not establish prediction. We tested whether instability computed on $N-1$ cell lines predicts whether a drug's effect on the held-out Nth cell line will be consistent, using leave-one-out cross-validation across 2,694 drugs with ≥ 10 cell lines.

For each drug and each cell line, we computed direction instability on the remaining $N-1$ cell lines (the LOO instability), then measured the cosine similarity between the held-out signature and the consensus of the remaining $N-1$. Because the held-out cell line never enters the instability calculation, any predictive relationship is genuinely out-of-sample.

LOO instability predicted held-out cosine with Spearman $\rho = -0.98$ ($p < 10^{-300}$): drugs with low instability (computed without seeing the held-out line) produce held-out signatures that closely match the consensus (Figure 6a). Binarizing held-out outcomes as "consistent" (cosine > 0.3) versus "inconsistent," LOO instability achieved AUROC = 0.986. This high value partly reflects shared cosine geometry between the predictor (D, built from pairwise cosines) and the outcome (held-out cosine with consensus); the permutation control in Section 3.3 shows that the relationship depends on genuine drug-level structure rather than geometric semi-circularity. The separation between the most stable quartile (Q1) and least stable quartile (Q4) is nearly complete (Figure 6b): stable drugs produce held-out cosines of 0.2–0.7, while unstable drugs cluster below 0.2.

The HDAC selectivity gradient survived in the predictive setting (Table 3). Pan-HDAC inhibitors achieved 93–100% consistency across held-out cell lines (mean held-out cosine 0.63–0.71), while selective inhibitors achieved 0–31% consistency (mean cosine 0.08–0.25). Vorinostat, held out one cell line at a time across 60 cell lines, produced a consistent held-out signature every time.

Robustness controls. The AUROC of 0.986 raises a natural concern: is the held-out prediction structurally trivial? We performed a permutation test, shuffling the mapping between drugs and their instability values 1,000 times and recomputing AUROC and ρ each time. The permuted AUROC collapsed to 0.500 ± 0.012 (Figure 7a), and the permuted ρ collapsed to -0.001 ± 0.019 (Figure 7b). No permutation achieved an AUROC ≥ 0.986 or $|\rho| \geq 0.976$ (empirical $p < 0.001$; 0 of 1,000). The predictive relationship depends on genuine drug-level structure, not on the statistical architecture of the LOO procedure.

A second concern is whether cross-cell-line variation reflects biology or assay noise. We computed a replicate instability D_{rep} for each drug: the mean pairwise angular deviation among Level 5 signatures sharing the same drug, cell line, dose, and timepoint (i.e., true replicates differing only by plate). D_{rep} provides an empirical noise floor—cross-cell instability is biologically meaningful only when it exceeds D_{rep}. Across 1,515 drugs with ≥ 2 replicate groups, cross-cell D exceeded D_{rep} for 68% of drugs (Figure 8a), with a

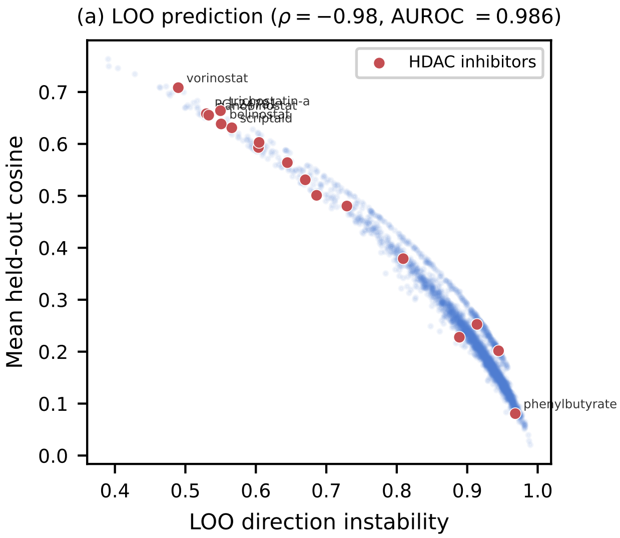

Figure 6: Held-out prediction of drug mechanism transport. **(a)** Direction instability computed on $N-1$ cell lines (x-axis) versus mean cosine between the held-out signature and the $N-1$ consensus (y-axis), for 2,694 drugs with ≥ 10 cell lines. Red points are HDAC inhibitors. The tight curve ($\rho = -0.98$, AUROC $= 0.986$) shows that instability *predicts* held-out consistency, not merely correlates with it. **(b)** Histogram of held-out cosine for the most stable (Q1, blue) and least stable (Q4, red) quartiles. The distributions barely overlap: stable drugs reliably transport to new contexts.

Table 3: Held-out prediction for HDAC inhibitors. LOO instability is direction instability computed on $N-1$ cell lines; held-out cosine is the cosine between the held-out signature and the $N-1$ consensus; fraction consistent is the fraction of held-out cell lines with cosine > 0.3.

Drug	n	LOO D	Held-out cos	Frac. cons.
vorinostat	60	0.490	0.708	1.00
panobinostat	15	0.533	0.655	1.00
trichostatin-a	64	0.550	0.664	1.00
belinostat	13	0.551	0.638	1.00
scriptaid	15	0.566	0.631	0.93
entinostat	14	0.604	0.593	0.93
PCI-34051	12	0.889	0.228	0.25
givinostat	13	0.914	0.253	0.31
valproic acid	54	0.944	0.202	0.26
phenylbutyrate	11	0.968	0.081	0.00

median ratio of 1.04. The HDAC panel (Figure 8b) makes the gradient concrete: pan-isoform inhibitors produce highly reproducible replicates ($D_{rep} = 0.17$–0.49) and low cross-cell instability ($D = 0.59$–0.63), while selective inhibitors are noisy within-cell ($D_{rep} = 0.82$–0.98) and noisy across cells ($D = 0.70$–0.98). The selectivity gradient appears in both metrics because the underlying mechanism differs: pan-inhibitors target ubiquitous chromatin machinery and produce clean, directionally consistent signatures everywhere, while selective inhibitors produce strong signals only where their isoform-specific target is functionally relevant.

Direction instability also correlates with mean signature L_2 norm ($\rho = -0.41$): drugs with stronger signals tend to be more stable. Signal strength predicts held-out cosine ($\rho = 0.59$), but instability explains 1.7×

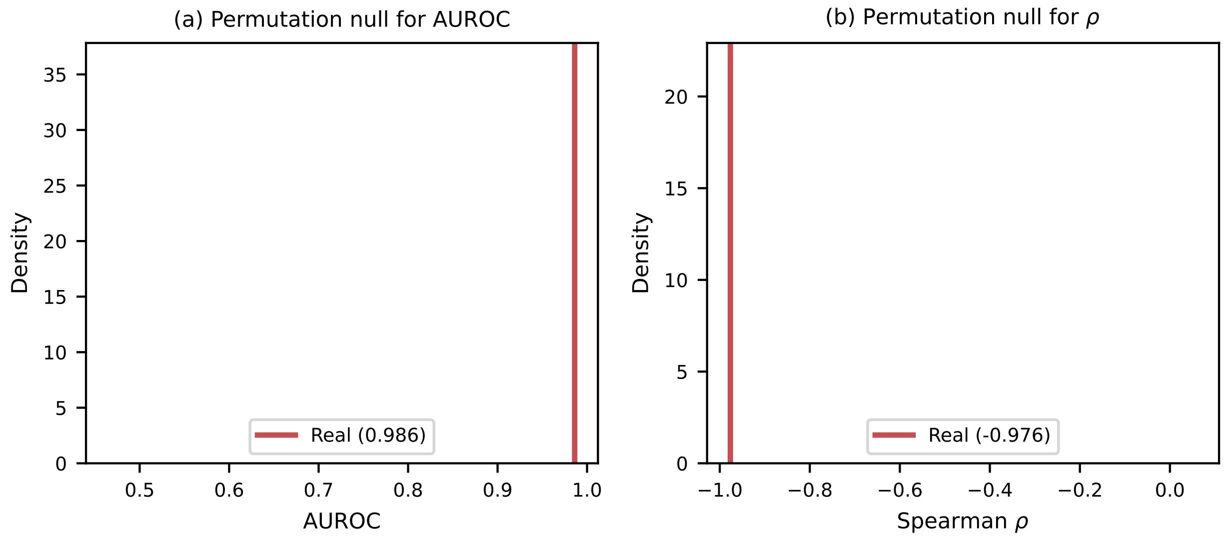

Figure 7: Permutation control for held-out prediction. **(a)** Distribution of AUROC under 1,000 random permutations of drug-instability assignments. The real AUROC (0.986, red line) is entirely separated from the null distribution (mean 0.500). **(b)** Same permutation test for Spearman ρ. The real correlation (-0.976) falls outside the entire null distribution.

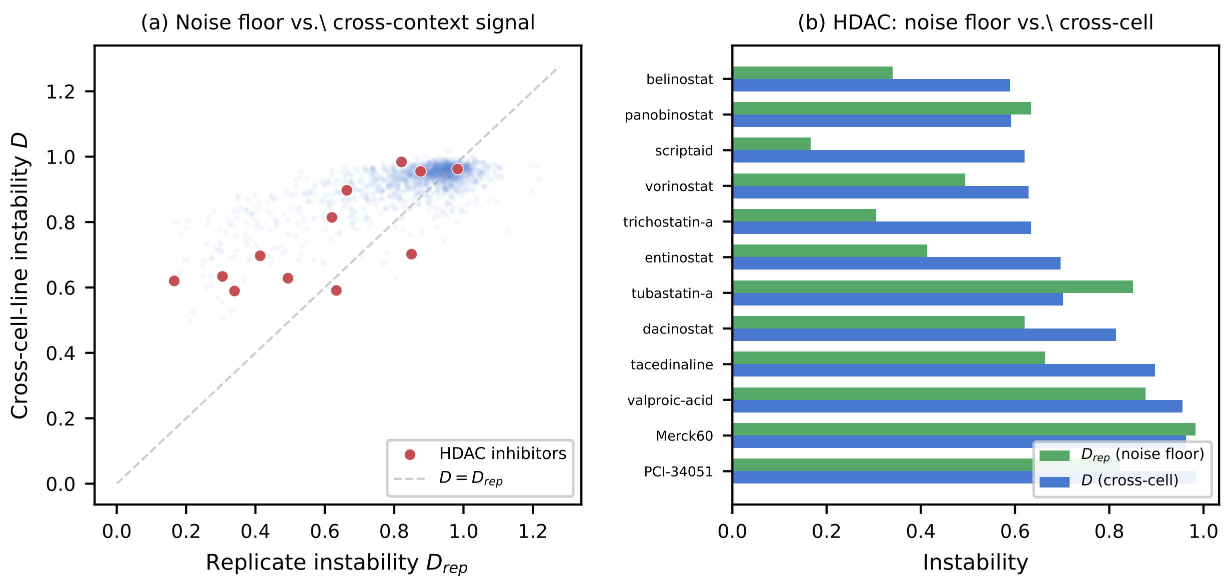

Figure 8: Replicate noise floor. **(a)** Replicate instability D_{rep} (within-cell, same-dose, same-timepoint) versus cross-cell-line instability D for 1,515 drugs with ≥ 2 replicate groups. Points above the dashed identity line have cross-cell variation exceeding their replicate noise floor. HDAC inhibitors (red) span from low-noise, low-instability (pan-isoform) to high-noise, high-instability (selective). **(b)** HDAC panel: D_{rep} (green) and cross-cell D (blue) for each inhibitor. Pan-isoform inhibitors are both reproducible and consistent; selective inhibitors are noisy everywhere.

more variance ($|\rho| = 0.98$ vs. 0.59). Signal quality is a partial contributor to directional consistency, as expected—poorly measured signatures produce noisy directions—but instability captures structure beyond what signal strength accounts for.

4 Discussion

Direction instability provides a single-number summary of whether a drug's transcriptomic mechanism generalizes across cell types. Three lines of evidence support this claim. First, the metric tracks gene-level mechanism conservation ($\rho = -0.79$ with Jaccard overlap), MOA classes stratify in a biologically interpretable pattern, and the cytotoxicity defense shows that the signal reflects target-specific conservation rather than generic cell death. Second, the HDAC selectivity gradient provides a within-class controlled comparison that rules out MOA labels as the driver: target breadth determines transport. Third, leave-one-out cross-validation elevates the claim from correlation to prediction (AUROC = 0.986, permutation collapse to 0.500), with a replicate noise floor showing that cross-cell variation exceeds assay noise for the majority of drugs.

The transporting-core analysis identifies the specific genes that are conserved: for pan-HDAC inhibitors, the shared core is the HDAC target pathway itself (*SUV39H1*, *MYC*, *CDK6*). Genetic triangulation provides suggestive convergent evidence from an independent data modality: within five drug families, exploratory within-family analysis found that mechanism-conserving drugs produce signatures more similar to genetic knockdown of their annotated targets, though the pre-registered global test fell short of its effect-size threshold. Independent replication of the within-family result would strengthen the case that transport is target-mediated.

Relation to the bracket norm. Direction instability is formally analogous to the neural bracket norm (Tower, 2026), which measures directional rotation of learned representations across evidence levels. Both metrics detect when a system responds to a perturbation by rotating its internal state rather than scaling it. The pharmacological application replaces evidence quartiles with cell lines and neural activations with gene expression signatures, preserving the geometric interpretation: low rotation implies a conserved mechanism that transports across contexts. The analogy is structural, not literal: evidence quartiles are ordered while cell lines are exchangeable, so the bracket norm's monotonicity properties do not carry over. Direction instability treats all pairwise comparisons symmetrically, which is appropriate for the unordered cell-line setting.

Honest nulls. Three pre-registered predictions missed their thresholds. The global genetic triangulation correlation ($\rho = -0.18$) fell below the pre-registered bar of $|\rho| > 0.3$; the within-family stratification that rescued this result was not pre-registered. Component-tissue enrichment showed no signal (0.6% vs. the predicted >20%). The strict component-count prediction for pan-HDAC inhibitors (1–2 components) was too tight (observed 1–5). All three are reported because the pre-registration commit SHA proves they were genuine predictions, not post-hoc hypotheses.

Falsifiability. The central claim—that direction instability tracks mechanism conservation—would be falsified by a drug class with a known conserved mechanism that scores as unstable, or by showing that the held-out prediction collapses on an independent perturbation dataset (JUMP-CP, Perturb-seq). The HDAC selectivity gradient would be undermined if isoform-selective inhibitors produced conserved signatures in a panel enriched for cell types expressing the relevant isoform. We have found no such counterexample in LINCS L1000, but absence of counterexamples in one dataset is weaker than replication across datasets.

Limitations. LINCS L1000 measures only 978 landmark genes; full-transcriptome profiling might reveal additional structure. The population null controls for cell-line count but not for other confounds (drug concentration, time point). Direction instability partially correlates with signature L_2 norm ($\rho = -0.41$) and with replicate instability ($\rho = 0.51$): drugs with stronger, more reproducible signals tend toward lower cross-cell instability. Cross-cell D exceeds the replicate noise floor for 68% of drugs and instability explains 1.7× more held-out variance than norm alone, so the metric captures structure beyond assay noise, but the two are partially confounded. A Level 4 (pre-replicate-collapse) analysis would provide a tighter noise bound than the plate-level replicates used here. MOA annotations from the Repurposing Hub are incomplete (20% coverage) and may contain errors. The direction-instability metric treats all cell lines as exchangeable and does not model phylogenetic relationships between cell types.

Applications. Direction instability could serve as a filter for identifying compounds whose transcriptomic mechanism is conserved across cellular contexts, a prerequisite for robust cross-cell-line translation. Whether cell-line transport predicts clinical efficacy across patient populations remains an open question outside the scope of this work. The metric requires only perturbation signatures across ≥ 5 contexts, making it applicable to any multi-context profiling dataset (CMAP, JUMP-CP, Perturb-seq).

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